

Question 1. Find a vector $\vec{v} \in \mathbb{R}^3$ that satisfies each of the following conditions.

- (1) $\vec{v}$ points from $(0, 1, 2)$ to $(1, 1, 1)$. (2) $\vec{v}$ is the projection of $\langle 1, 2, 3 \rangle$ onto $\langle 1, 1, 1 \rangle$.
- (3) $\vec{v}$ is a unit vector in the plane $x - 2y + z = 0$. (4) $\vec{v} \neq 0$ is in the intersection of the plane $x - 2y + z = 0$ and the plane $2x + y - z = 0$.
- (5) $\vec{v} \neq 0$ is a vector orthogonal to $\langle 1, -2, 1 \rangle$ and to $\langle 2, 1, -1 \rangle$. (6) $\vec{v} \neq 0$ is a vector in the plane $x + y + 2z = 0$ and is orthogonal to $\langle 1, -1, 1 \rangle$.
- (7) $\vec{v}$ is the projection of $\langle 1, 2, 3 \rangle$ onto the plane $x + y + z = 0$. (8) (challenge) $\vec{v}$ is a unit vector in the plane $z = 0$ and forms an angle of $\frac{\pi}{4}$ with $\langle 3, 4, 5 \rangle$.

Question 2. Find the area or volume of the following shapes in $\mathbb{R}^3$.

- (1) The area of the triangle with vertices $(0, 1, 0)$, $(1, 1, -1)$, and $(1, 2, 3)$.
- (2) The volume of the parallelepiped with a vertex at the origin with adjacent vertices at $(2, 1, -1)$, $(1, 1, 0)$, $(1, 1, 1)$.